

Exam Corrections — Model Answers

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Award marks for any answer demonstrating correct understanding, even if wording differs. Grade = (points / 40) × 20.

Q1a (4 pts) — FRic–richness confound

The conclusion is not fully valid. FRic mechanically increases with species richness because adding species to a trait space tends to expand its volume, even if the new species are functionally redundant. The correlation may simply reflect this mathematical artefact rather than genuine ecological differences in functional organisation.

Award: identifies confound (2 pts), explains mechanism (2 pts).

Q1b (6 pts) — SES-FRic

Compute the **standardised effect size (SES-FRic)**. Build a null model by randomly shuffling species among communities (keeping richness constant) and compute FRic for each randomisation. $SES = (\text{observed FRic} - \text{mean null FRic}) / \text{sd null FRic}$. $SES > +1.96$ ($p < 0.05$): genuinely overdispersed. SES removes the richness effect — whatever remains is a non-random assembly signal.

Award: names SES or null model (2 pts), explains logic (2 pts), threshold (2 pts).

Q2a (5 pts) — FRic interpretation

Both communities have the same number of species ($S = 12$) and same dominant trait ($CWM = 8$ cm). Community 1 ($FRic = 0.70$): species span a wide range of body sizes — the functional space is well filled. Community 2 ($FRic = 0.15$): all 12 species are clustered in a narrow region of trait space around 8 cm — very similar body sizes, little functional diversity.

Award: correct interpretation of both (3 pts), notes same CWM but different FRic (2 pts).

Q2b (5 pts) — Mechanism for low FRic

Acceptable mechanisms (any one, well-argued): (1) **Environmental filtering**: a harsh/uniform environment (e.g. fast cold stream) only allows species with a narrow range of body plans. (2) **Recent colonisation**: a young community hasn't yet diversified — few lineages, functionally similar. (3) **Strong predation** selecting for specific body sizes. Key: mechanism must logically cause convergence of traits.

Award: valid mechanism (3 pts), logical link to FRic (2 pts).

Q3a (5 pts) — Neotropical overdispersion

$SES \approx +2.6$ (above $+1.96$) = **functional overdispersion**: assemblages occupy more functional space than expected by chance. Assembly process: **competitive exclusion / niche differentiation**.

Mechanism: in species-rich Amazonian rivers, coexisting species have diverged to reduce competition — morphologically distinct species partition resources (benthic vs pelagic, large vs small mouth, etc.).

Award: overdispersion identified (2 pts), competitive exclusion / niche partitioning (3 pts).

Q3b (5 pts) — Palearctic clustering

SES ≈ -1.8 (approaching -1.96) = tendency toward **functional clustering**: assemblages are less functionally diverse than expected. Assembly process: **environmental filtering**. Mechanism: cold, glacially impacted, often degraded Palearctic rivers filter out species with extreme functional strategies — only species with similar, cold-tolerant, streamlined body plans can persist.

Award: clustering / filtering identified (2 pts), valid mechanism (3 pts).

Q4 (10 pts) — Why FD matters for conservation

Key points to reward (2–3 pts each, up to 10 total):

- Species richness treats all species as equivalent, ignoring their ecological roles. Two communities can have the same S but very different trait compositions. (2 pts)
- Functional diversity predicts ecosystem functioning: traits of dominant species (CWM) and range of strategies (FRic) drive nutrient cycling, productivity, resilience to disturbance. Loss of FD = loss of functions even if S is maintained. (3 pts)
- Threatened species are often functionally unique (Carmona et al. 2021): their loss causes irreplaceable gaps in the functional space. Non-native species cannot substitute. (3 pts)

• Freshwater fish example: Neotropical realm hosts >75% of global fish functional diversity (Toussaint et al. 2016) — protecting Amazonian rivers is disproportionately important for global functional diversity, not just species numbers. (2 pts)

Award flexibly. Full marks require: S richness critique + FD–function link + example.

Total: 40 pts → grade /20. Suggested curve: 36-40=18-20 · 28-35=14-17 · 20-27=10-13 · <20=<10